Tests for one or two means

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```
bookdown::render_book()
```

Overview

Topics:

1. Random Sampling
2. Randomization
3. Tests for proportions
4. Tests for means
5. Exercises

Chapter 1

Randomization and Random Samples

Is a sample statistic statistically higher or lower than a population mean? **We need a random sample in this scenario.** Acock (2023) uses the example of a small liberal art college about selectivity compared to the population. The mean score for a small liberal arts college for math and reading was $\bar{x} = 1180$ while the population average SAT score is $\mu = 1080$. Can the liberal arts school say that there selectivity is higher than the average college? We can test the example above with a *one-group test for means or proportions*.

Does a treatment has an impact on the treatment group? **We need randomization into treatment and control in this scenario.** Acock (2023) uses the example of a reading program for preschool children. There is a randomization of preschool students in and out of the reading program. On a outcome of 0-100 point scale, the children in the control group had an average score of $\bar{y} = 71.3$ with a standard deviation of $sd = 10.4$. In the treatment group, the average score was $\bar{y} = 82.5$ and $sd = 9.9$. It seems like the reading program had an impact, but we would need to test for statistical significant. We can use a *two-group test for means or proportions* for this example.

1.1 Random Samples

Stata makes random samples and randomization straightforward and easy. We can implement random sampling without replacement with the *sample* command. Recall that sampling without replacement leads to changes the probability of being selected as the available pool declines.

```
clear
set obs 30
generate id = _n
list
set seed 54613
sample 15, count
list
```

```
Number of observations (_N) was 0, now 30.
```

```
     +----+
     | id |
     |----|
  1. |  1 |
  2. |  2 |
  3. |  3 |
  4. |  4 |
  5. |  5 |
     |----|
  6. |  6 |
  7. |  7 |
  8. |  8 |
  9. |  9 |
 10. | 10 |
     |----|
 11. | 11 |
 12. | 12 |
 13. | 13 |
 14. | 14 |
 15. | 15 |
     |----|
 16. | 16 |
 17. | 17 |
 18. | 18 |
 19. | 19 |
 20. | 20 |
     |----|
 21. | 21 |
 22. | 22 |
 23. | 23 |
 24. | 24 |
 25. | 25 |
     |----|
 26. | 26 |
 27. | 27 |
```

```
 28. | 28 |
 29. | 29 |
 30. | 30 |
     +----+


(15 observations deleted)

     +----+
     | id |
     |----|
  1. | 16 |
  2. |  2 |
  3. |  9 |
  4. | 13 |
  5. | 24 |
     |----|
  6. | 12 |
  7. | 23 |
  8. | 25 |
  9. |  6 |
 10. | 20 |
     |----|
 11. | 28 |
 12. | 21 |
 13. | 26 |
 14. | 29 |
 15. |  1 |
     +----+
```

We can try this exercise again but with a different seed.

```
clear
set obs 30
generate id = _n
set seed 11111
sample 15, count
list
```

```
Number of observations (_N) was 0, now 30.



(15 observations deleted)
```

```
     +----+
     | id |
     |----|
  1. | 21 |
  2. | 14 |
  3. | 19 |
  4. | 30 |
  5. | 15 |
     |----|
  6. |  3 |
  7. | 27 |
  8. | 18 |
  9. |  1 |
 10. | 29 |
     |----|
 11. | 16 |
 12. | 23 |
 13. | 20 |
 14. |  5 |
 15. |  2 |
     +----+
```

We can see that we come up with a new sample even though the only thing we changed was the set seed number.

1.2 Randomization and random assignment

We covered taking a random sample from our data, but how do we randomly assign in Stata. There are many ways to implement random assignment, but we'll cover a simple way to do so. Let's say we have pre-SAT scores before treatment with $\mu = 1000$ and $\sigma = 50$. We randomly assign into two groups for a program to prepare for the SAT. We will use the following code with the function *runiform()* to randomly assign student and and out of treatment.

```
clear
set obs 100
set seed 101010
gen id = _n
gen preSAT = rnormal(1000,50)
sum preSAT, detail
gen random_num = runiform()
sort random_num
gen rank = _n
```

```
gen treatment = 1 if rank <= 50
list if treatment == 1
```

```
Number of observations (_N) was 0, now 100.
```

```
                          preSAT
-------------------------------------------------------------
      Percentiles      Smallest
 1%     871.3848       864.4878
 5%     909.6474       878.2817
10%     929.7544       898.4756       Obs                 100
25%     960.7722       900.3383       Sum of wgt.         100

50%     996.7658                      Mean            994.422
                        Largest       Std. dev.       50.15997
75%     1026.011        1078.68
90%     1057.898        1082.178      Variance        2516.022
95%     1070.932        1082.615      Skewness        .0148653
99%     1118.523        1154.43       Kurtosis        3.25588
```

```
(50 missing values generated)
```

```
     +-----------------------------------------------+
     | id     preSAT   random~m   rank   treatm~t |
     |-----------------------------------------------|
  1. |  6   953.2567   .0166816      1          1 |
  2. | 75   1027.424   .0174284      2          1 |
  3. | 21   947.8138   .0601821      3          1 |
  4. | 64   964.9872   .0711294      4          1 |
  5. | 92   1051.042     .07504      5          1 |
     |-----------------------------------------------|
  6. | 22    1154.43   .0955792      6          1 |
  7. | 98   938.0862   .0960075      7          1 |
  8. | 46   973.4328   .1364803      8          1 |
  9. | 54   1014.777   .1383776      9          1 |
 10. | 19   1025.269   .1552295     10          1 |
     |-----------------------------------------------|
 11. | 67   1016.115   .1552606     11          1 |
 12. | 16   986.5502   .1599387     12          1 |
 13. |  9   878.2817   .1856076     13          1 |
 14. | 36   898.4756   .1968061     14          1 |
 15. | 87   918.5071   .2096984     15          1 |
```

```
     |-----------------------------------------------|
 16. |  8   972.4597   .2128657     16          1 |
 17. | 14   951.2537   .2144214     17          1 |
 18. | 12   963.9334   .2309978     18          1 |
 19. | 27   972.9071   .2562377     19          1 |
 20. | 45   961.1827    .266596     20          1 |
     |-----------------------------------------------|
 21. | 38    1054.86   .2714473     21          1 |
 22. | 29   941.5311   .2721756     22          1 |
 23. | 40   928.1702   .2802632     23          1 |
 24. | 24   964.6517   .2998621     24          1 |
 25. | 56   973.2595    .302796     25          1 |
     |-----------------------------------------------|
 26. | 59   1015.051   .3046456     26          1 |
 27. | 42   977.6315   .3150471     27          1 |
 28. | 25   1003.752   .3170995     28          1 |
 29. | 47   1032.023   .3493558     29          1 |
 30. | 73   864.4878   .3589761     30          1 |
     |-----------------------------------------------|
 31. | 60   1022.622   .3599698     31          1 |
 32. | 84   980.5345   .3756694     32          1 |
 33. | 76   947.2341   .3840492     33          1 |
 34. | 91   995.9388   .4091457     34          1 |
 35. | 55   1020.875   .4218919     35          1 |
     |-----------------------------------------------|
 36. | 44   1005.222   .4256864     36          1 |
 37. | 61   960.3616   .4356995     37          1 |
 38. | 23   929.7324   .4378634     38          1 |
 39. |  3   1012.586   .4518479     39          1 |
 40. | 20   1054.404   .4530961     40          1 |
     |-----------------------------------------------|
 41. | 90   967.8745    .453654     41          1 |
 42. | 69   988.4484    .475242     42          1 |
 43. | 78   979.8333   .4932367     43          1 |
 44. |  2   1055.188    .495618     44          1 |
 45. | 72   1018.799   .4993747     45          1 |
     |-----------------------------------------------|
 46. | 74    1078.68   .5066093     46          1 |
 47. | 95   1022.446   .5149328     47          1 |
 48. | 10   1043.046   .5175314     48          1 |
 49. |  5   1031.013   .5386587     49          1 |
 50. | 43   974.0781    .553867     50          1 |
     +-----------------------------------------------+
```

Let's check pre-SAT scores after randomized assignment.

```
bysort treatment: sum preSAT
```

```
-> treatment = 0

    Variable |        Obs        Mean    Std. dev.       Min        Max
-------------+---------------------------------------------------------
      preSAT |         50    1000.554    47.79832   900.3383   1082.615

-------------------------------------------------------------------------------------------------------------
-> treatment = 1

    Variable |        Obs        Mean    Std. dev.       Min        Max
-------------+---------------------------------------------------------
      preSAT |         50    988.2904    52.17265   864.4878    1154.43
```

Our *summarize* command helps show that the treatment and control group have similar pre-SAT scores. However, we will discuss two-group tests to test the null hypothesis that these two groups are similar.

1.3 Significance level of $p = 0.05$

I want to focus on Acock's (2023) discussion of the significance level of $p = 0.05$. We test our null hypothesis, H_0 with a one-tailed or two-tailed test. How do we know when to reject H_0? The rule of thumb for statistical analysis has been when $p < 0.05$. We say that we reject the null hypothesis at the 5% level. Some researchers use $p < 0.01$ to say that we reject the null hypothesis at the 1% level or $p < 0.001$ to say that we reject the null hypothesis at the 0.1% level.

We may find $p < 0.05$ just due to the fact that our N is extremely large. We may find statistical significance, but is it practical? We may find $p < 0.05$ but our assumptions have been violated. (This is a preview for Econ 645 and Econ 672). Therefore, we risk rejecting a true null beyond our 5% level due to the bias. We may find $p < 0.05$ if try to data mine and cherry pick results. Economist try to prevent cherry picking with extensive sensitivity and robustness tests. (Have you seen the length of online appendices?)

Here are some of my thoughts on $p < 0.05$

1. Always have a healthy skepticism that your assumptions have been satisfied.
2. Think about the arbitrariness of $p = 0.051$ and $p = 0.049$.
3. Don't cherry pick results. Run sensitivity tests to try to break your results.

Chapter 2

Tests for proportions and means

We can use a one-sample test when we want to compare a sample mean to a population mean or a sample proportion to a population proportion. We can use a two-sample test of groups to compare two samples.

2.1 One-sample test of proportions

Acock compares adult opinions on school prayer from the 2006 General Social Survey. We will use the *recode* command to convert 2 into 0 and have a binary $[0, 1]$ variable. We will use the *tabulate* command to make sure our recode worked.

```
cd "/Users/Sam/Desktop/Econ 643/Data/agis6r/"
use "gss2002_chapter7.dta", clear
recode prayer (1 = 1 Approve) (2 = 0 Disapprove), gen(schpray)
tabulate schpray prayer, missing
```

```
/Users/Sam/Desktop/Econ 643/Data/agis6r


(518 differences between prayer and schpray)


 RECODE of |
    prayer |
    (bible |
```

```
 prayer in |
    public |  bible prayer in public schools
  schools) |   approve  disapprov          . |     Total
-----------+---------------------------------+----------
Disapprove |         0        518          0 |       518
   Approve |       341          0          0 |       341
         . |         0          0      1,906 |     1,906
-----------+---------------------------------+----------
     Total |       341        518      1,906 |     2,765
```

Next we will test the null hypothesis that half of the people approve of school prayer.

$$H_0 : p = 0.5$$

$$H_A : p \neq 0.5$$

We will test this hypothesis with the *prtest* command with a two-tailed test. This will do a test of proportions for a one-group sample. The one-sample test of proportion compares a sample to a defined population mean.

```
prtest schpray == 0.5
```

```
One-sample test of proportion                  Number of obs      =       859

------------------------------------------------------------------------------
    Variable |       Mean   Std. err.                     [95% conf. interval]
-------------+----------------------------------------------------------------
     schpray |   .3969732   .0166937                      .3642542    .4296923
------------------------------------------------------------------------------
    p = proportion(schpray)                                       z =  -6.0392
H0: p = 0.5

     Ha: p < 0.5                 Ha: p != 0.5                 Ha: p > 0.5
 Pr(Z < z) = 0.0000         Pr(|Z| > |z|) = 0.0000          Pr(Z > z) = 1.0000
```

Our sample mean is $\bar{p} \approx 0.4$. We reject the null hypothesis that half of the population approves of school prayer at the 0.1% level.

2.2 Two-sample tests of proportions

We may want to compare proportions between two samples. Our prior example used SAT scores as an outcome of interest, where the treatment was a SAT

preparation program. We saw that pre-treatment pre-SAT scores were similar between treatment and control. Now we want to compare the proportion of students who scored above $1,200$.

Let's look at our treatment and control again.

```
tab treatment
```

```
  treatment |      Freq.     Percent        Cum.
------------+-----------------------------------
          0 |         50       50.00       50.00
          1 |         50       50.00      100.00
------------+-----------------------------------
      Total |        100      100.00
```

We can check the null hypothesis that the proportion of treatment students, who scored higher than a $1,200$, was the same proportion as control students.

$$H_0 : p_{treatment} - p_{control} = 0$$
$$H_A : p_{treatment} - p_{control} \neq 0$$

Our data are in a long-format, so we will use the *by* option with the *prtest* command.

```
list treatment post1200 in 46/55
tabulate post1200 treatment
prtest post1200, by(treatment)
```

```
     | treatm~t   post1200 |
     |---------------------|
 46. |        1          1 |
 47. |        1          0 |
 48. |        1          0 |
 49. |        1          1 |
 50. |        1          0 |
     |---------------------|
 51. |        0          0 |
 52. |        0          0 |
 53. |        0          0 |
 54. |        0          0 |
 55. |        0          0 |
     +---------------------+

           |       treatment
```

```
  post1200 |         0          1 |     Total
-----------+----------------------+----------
         0 |        44         25 |        69
         1 |         6         25 |        31
-----------+----------------------+----------
     Total |        50         50 |       100
```

```
Two-sample test of proportions                     0: Number of obs =       50
                                                   1: Number of obs =       50
------------------------------------------------------------------------------
       Group |       Mean   Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
           0 |        .12   .0459565                      .0299269    .2100731
           1 |         .5   .0707107                      .3614096    .6385904
-------------+----------------------------------------------------------------
        diff |       -.38   .0843327                     -.545289    -.214711
             |  under H0:   .0924986    -4.11   0.000
------------------------------------------------------------------------------
        diff = prop(0) - prop(1)                                  z =  -4.1082
    H0: diff = 0

    Ha: diff < 0                 Ha: diff != 0                 Ha: diff > 0
 Pr(Z < z) = 0.0000         Pr(|Z| > |z|) = 0.0000          Pr(Z > z) = 1.0000
```

We see that a higher proportion of treatment student scored higher than $1,200$. We reject the null hypothesis at the 0.01% level using a two-tailed test of proportions.

Please note that Acock (2023) uses an example of *prtest* when the data are in a wide format. You can review this, but your data should be organized in a long-format.

2.3 One-Sample test of means

We can use Stata to conduct t-tests for one-sample or two-sample tests of means. Our main command is *ttest*. For our one-sample test of means, we will compare the sample mean to a population mean. We hypothesize that the population mean of working hours for employed individuals is 40 or $\mu = 40$. Let's find the sample mean of working hours from the 2006 General Social Survey.

```
cd "/Users/Sam/Desktop/Econ 643/Data/agis6r/"
use "gss2002_chapter7.dta", clear
summarize hrs1 if wrkstat==1
```

```
/Users/Sam/Desktop/Econ 643/Data/agis6r

    Variable |        Obs        Mean    Std. dev.       Min        Max
-------------+---------------------------------------------------------
        hrs1 |      1,419    45.97111    11.88977          4         89
```

We can see that $\bar{x} = 46$ and $s = 11.9$. Let's test to see if this is statistically significant different from the hypothesis of a population mean of 40.

$$H_0 : \mu_{(hours)} = 40$$

$$H_A : \mu_{(hours)} \neq 40$$

```
ttest hrs1==40 if wrkstat==1
```

```
One-sample t test
------------------------------------------------------------------------------
Variable |     Obs        Mean    Std. err.   Std. dev.   [95% conf. interval]
---------+--------------------------------------------------------------------
    hrs1 |   1,419    45.97111    .3156328    11.88977    45.35195    46.59026
------------------------------------------------------------------------------
    mean = mean(hrs1)                                             t =  18.9179
H0: mean = 40                                  Degrees of freedom =     1418

    Ha: mean < 40               Ha: mean != 40                 Ha: mean > 40
 Pr(T < t) = 1.0000         Pr(|T| > |t|) = 0.0000          Pr(T > t) = 0.0000
```

We find that $t = 18.9179$ and $p < 0.001$. We reject the null hypothesis at the 0.01% level using a two-tailed t-test that the population mean. In addition, Stata provides a 95% confidence interval of $[45.3, 46.6]$. We find that the average working hours is greater than our hypothesized 40 work week.

2.4 Two-Sample test of group means

We can compare the means between two samples using the *ttest* command with the *by* option. Acock (2023) uses the example of earnings from men and women. We'll use the 2006 General Social Survey to assess the group means. Let's take a look at the income data first.

```
tabulate rincom98
```

```
    respondents |
         income |      Freq.     Percent        Cum.
----------------+-----------------------------------
   under $1 000 |         44        1.59        1.59
$1 000 to 2 999 |         69        2.50        4.09
$3 000 to 3 999 |         37        1.34        5.42
$4 000 to 4 999 |         24        0.87        6.29
$5 000 to 5 999 |         37        1.34        7.63
$6 000 to 6 999 |         20        0.72        8.35
$7 000 to 7 999 |         26        0.94        9.29
$8 000 to 9 999 |         60        2.17       11.46
$10000 to 12499 |         78        2.82       14.29
$12500 to 14999 |         85        3.07       17.36
$15000 to 17499 |         69        2.50       19.86
$17500 to 19999 |         86        3.11       22.97
$20000 to 22499 |        100        3.62       26.58
$22500 to 24999 |         81        2.93       29.51
$25000 to 29999 |        140        5.06       34.58
$30000 to 34999 |        166        6.00       40.58
$35000 to 39999 |        123        4.45       45.03
$40000 to 49999 |        170        6.15       51.18
$50000 to 59999 |        136        4.92       56.09
$60000 to 74999 |         95        3.44       59.53
$75000 - $89999 |         41        1.48       61.01
$90000- $109999 |         23        0.83       61.84
  $110 000 over |         70        2.53       64.38
        refused |        985       35.62      100.00
----------------+-----------------------------------
          Total |      2,765      100.00
```

What is the problem? We don't know the exact values. We only know the range of the values. Acock (2023) takes the midpoint, but this method is ripe with measurement error. Please see page 168 on how he recodes the variable.

Let's compare hours instead.

$$H_0 : \mu_{(men)} - \mu_{(women)} = 0$$

$$H_A : \mu_{(men)} - \mu_{(women)} \neq 0$$

```
ttest hrs if wrkstat==1, by(sex)
```

```
Two-sample t test with equal variances
------------------------------------------------------------------------------
```

```
   Group |     Obs        Mean    Std. err.   Std. dev.   [95% conf. interval]
---------+--------------------------------------------------------------------
    male |     748    48.08422    .4630884    12.66528    47.17512    48.99333
  female |     671     43.6155    .4045276    10.47875     42.8212    44.40979
---------+--------------------------------------------------------------------
Combined |   1,419    45.97111    .3156328    11.88977    45.35195    46.59026
---------+--------------------------------------------------------------------
    diff |            4.468725    .6211783                3.250197    5.687253
------------------------------------------------------------------------------
    diff = mean(male) - mean(female)                              t =   7.1939
H0: diff = 0                                     Degrees of freedom =     1417

    Ha: diff < 0                 Ha: diff != 0                 Ha: diff > 0
 Pr(T < t) = 1.0000         Pr(|T| > |t|) = 0.0000          Pr(T > t) = 0.0000
```

Our $t = 7.1939$ and $p < 0.001$. We reject the null hypothesis at the 0.01% level and conclude that men and women do not work the same amount of hours.

2.5 Testing for unequal variance

We can test to see if the variance or standard deviations are similar between two groups. Why do we care if the variance are equal? Because we assume that the variance is the same between two samples from the population. We can use the *sdtest* in Stata to test for equal variance. This test uses an F test for significant differences in variances.

The F test is a ratio of the variances between the two groups.

$$H_0 : ratio = 1$$

$$H_A : ratio \neq 1$$

```
sdtest hrs if wrkstat==1, by(sex)
```

```
Variance ratio test
------------------------------------------------------------------------------
   Group |     Obs        Mean    Std. err.   Std. dev.   [95% conf. interval]
---------+--------------------------------------------------------------------
    male |     748    48.08422    .4630884    12.66528    47.17512    48.99333
  female |     671     43.6155    .4045276    10.47875     42.8212    44.40979
---------+--------------------------------------------------------------------
Combined |   1,419    45.97111    .3156328    11.88977    45.35195    46.59026
```

```
------------------------------------------------------------------------------
    ratio = sd(male) / sd(female)                                 f =   1.4609
H0: ratio = 1                                 Degrees of freedom = 747, 670

    Ha: ratio < 1                 Ha: ratio != 1                 Ha: ratio > 1
  Pr(F < f) = 1.0000         2*Pr(F > f) = 0.0000           Pr(F > f) = 0.0000
```

We reject the null hypothesis and the variance between the two samples is not the same. This will matter when we focus on Analysis of Variance (ANOVA). In regression, one of assumption is constant variance across values or homoskedasticity.

Chapter 3

Exercises

3.1 Randomization

We will create a random sample from a random number generator.

Step

1. Create a blank file with 1000 observations
2. Generate an id variable based on the row number.
3. Set a seed to *91925*
4. Create a variable called *wages* with a random number generator with $\mu = 1500$ and $\sigma = 500$
5. Summarize *wages*. What is the mean, minimum, and maximum?
6. Generate a new random number called *random_num*.
7. Sort ascending using *random_um* and create a new ranking variable.
8. Create a new variable called treatment.
9. Those 500 and below are set at 0 and those at 501 to 1000 are set at 1
10. Label the treatment variable 0 "Control" 1 "Treatment".
11. Use a t-test to test the null hypothesis $\mu_{(control)} - \mu_{(treatment)} = 0$
12. What do you conclude?

3.2 National Longitudinal Survey of Youth

Do Latino families have more have more children than other ethnic groups? Load the file called "nlsy97_chapter7.dta" to answer the question.

Steps:

1. First summarize *hh18_97* which is the number of children under 18

2. Next, tabulate *ethnic97* which is a binary for Latino/Hispanic
3. Test the null hypothesis $H_0 : \mu_{(Latino)} - \mu_{(Non-Latino)} = 0$
4. What do you conclude?

Is the varaince the same for number of children for Latino families compared to other ethnic groups?

Steps:

1. First, summarize *hh18_97* which is the number of children under 18
2. Second, compare the standard deviation/variance by *ethnic97*, which is a binary for Latino/Hispanic
3. Test the null hypothesis $H_0 : \sigma_L^2 = \sigma_N^2$
4. What do you conclude?

3.3 General Social Survey

Are women more likely to say that abortion is okay under any circumstances than men? Load the file called "gss2002_chapter7.dta".

Steps

1. First, tabulate *abany*
2. Next, tabulate *sex*
3. Then do a cross-tabulation on *abany* and *sex*
4. Use the *prtest* command on *abany* and *sex*. What happened?
5. We need to recode our variable into a binary.
6. Generate a new variable based off of *abany* and label *No* 0 and *Yes* 1.
7. Finally, test the null hypothesis: $p_{(women)} - p_{(men)} - 1$
8. What do you conclude?

Acock, A.C. (2023). A gentle introduction to Stata. 6th ed. Stata Press.

Anderson, D. R., Sweeney, D. J., Williams, T. A., Camm, J. D., & Cochran, J. J. (2016). Statistics for business and economics (13th ed.). Cengage Learning.

Mitchell, M. N. (2020). Data management using Stata: A practical handbook. Stata Press.